LUIS ARTURO ALARCÓN BLANCO

Software Engineer | Developer · AI Integration

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PROFESSIONAL SUMMARY

Software Engineer / Developer with 5+ years of experience in web development, designing and delivering web applications, REST APIs and system integrations with Node.js, TypeScript, React and SQL databases (PostgreSQL, MySQL, SQL Server). Experienced across the full software development lifecycle: requirements analysis, software design (database structure, components, APIs and integrations), front-end and back-end development, unit testing and CI/CD deployment. Hands-on experience building AI-powered solutions, business process automation and AI-assisted development workflows. Has coordinated development teams through system implementations, with a strong focus on application performance, debugging and code quality.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, PHP, Java, SQL, Bash

Back-end & APIs: Node.js, Express.js, REST API design, Laravel, CodeIgniter, FastAPI, Spring Boot, PM2

Front-end: React, Next.js, Vite, Vue.js, Angular, HTML5, CSS3, Progressive Web Apps (PWA)

AI & Automation: AI integration, OpenAI Whisper, AI-assisted development, business process automation, cron jobs

Databases: PostgreSQL, MySQL, SQL Server, MongoDB, Firebase

Cloud & DevOps: AWS, Google Cloud Platform, Microsoft Azure, Docker, Linux, CI/CD, GitHub Actions

Testing & Quality: Unit testing, Jest, PHPUnit, Selenium, debugging, design patterns

Tools & Methods: Git, GitHub, GitLab, Jira, Trello, Asana, Figma, Agile

PROFESSIONAL EXPERIENCE

Full Stack Developer Oct 2023 – Oct 2026

Ciudad Maderas — Querétaro, México (on-site)

- Coordinated development teams during system implementations: broke down requirements into tasks, provided technical estimates, tracked progress and ensured on-time delivery.
- Integrated AI tools into the development workflow to accelerate code generation, code review, technical documentation and incident resolution, increasing team productivity.
- Designed and built an employee benefits platform with React, Next.js and Vite, including appointment scheduling (psychology and nutrition services), savings fund enrollment, corporate event registration and content dashboards for the Communications team.
- Developed and maintained the company CRM (PHP, Laravel, CodeIgniter) to automate business processes: sales tracking, customer follow-up, analytical reporting and administrative task automation.
- Resolved production incidents through debugging and root-cause analysis, delivering performance optimizations and improvements that kept the platform stable.

Full Stack Developer May 2022 – Oct 2023

Secretaría de Seguridad Pública de Querétaro (State Public Security Agency) — Querétaro, México (on-site)

- Built an AI speech recognition and storage system with OpenAI Whisper and Python to transcribe audio from police vehicle cameras and interpret non-standard police code language; automated processing with Bash scripts and cron jobs.
- Designed and developed API services with Node.js, Express.js and SQL Server (deployed with PM2) to integrate multiple applications and external platforms, enabling real-time data exchange, process automation and a spam phone number registry.
- Developed and maintained a police operations Progressive Web App (Node.js, Express.js) for CCTV monitoring, real-time GPS tracking of police units, traffic tablet management, crime incident visualization, 911 panic button monitoring and patrol route tracking.

- Developed and maintained a 911 emergency video call system (Laravel, Vue.js, Docker, Jitsi Meet) that displays the caller's data and situation in real time for people under protection orders.
- Built a document digitization system for Human Resources (Laravel, Vue.js, Docker, Google Cloud Platform) to manage personnel records and internal procedures.

Mobile DeveloperJun 2021 – Jun 2024
Metrología Integral y Desarrollo S.A. de C.V. — Querétaro, México (On-site)

- Developed a cross-platform ERP application (mobile and web) with React Native, PHP REST APIs and MySQL for real-time inventory control of calibration equipment, reducing operational costs.
- Built a mobile app for precise timing and registration of professional swimmers, with performance history, individual progress analytics and sports event management.

EDUCATION

B.Eng. in Software Development and Management Aug 2018 – Apr 2022
Universidad Tecnológica de Querétaro

Technical Diploma in Computer Science Jul 2015 – Jun 2018
Centro de Estudios Tecnológicos Industrial y de Servicios No. 16 (CETIS 16)

LANGUAGES

Spanish: Native **English:** Upper-Intermediate (B2)